

Linux Live Response Digital Forensics Collector

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PROJECT OVERVIEW

The Problem

When a system is compromised, responders need fast answers:

- Who is logged in?
- What processes are running?
- Has persistence been established?

Doing this manually is slow, error-prone, and hard to document consistently.

Our Solution

A Bash-based forensics tool that:

- Runs on a live Linux system
- Collects evidence read-only (no tampering)
- Detects attacker persistence mechanisms
- Produces a consistent, defensible report

ARTIFACT COLLECTION

System Info	Users & Auth	Processes	Network	Log Files
• Kernel & OS version • Hostname & timezone • CPU & memory	• Who is logged in • Login history • Bash history files	• Running processes • Kernel modules • Active systemd services	• Active connections • Listening ports • Routing table	• Auth logs • System logs • Journal logs

PERSISTENCE DETECTION

The tool scans for common techniques attackers use to survive a reboot:

1. Startup Scripts

Checks `/etc/rc.local` and `/etc/init.d/` for unauthorized entries

2. Cron Jobs

Scans both user-level and system-wide scheduled tasks

3. Systemd Services & Timers

Lists and flags suspicious services or timers

4. SUID/SGID Binaries

Finds files with elevated permissions that attackers abuse

HOW IT WORKS

1

Run main.sh

Entry point — starts the collection process with a timestamp

2

collector.sh

Gathers all baseline artifacts (system, users, processes, network, logs)

3

Modules

Specialized scripts in /modules/ handle each category of data

4

Output

Results saved to /output/ — organized, timestamped, ready for review

Thank You

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